Intelligent Systems with Applications 26 (2025) 200514

Contents lists available at ScienceDirect

Intelligent Systems with Applications

journal homepage: www.journals.elsevier.com/intelligent-systems-with-applications

Explainable AI based LightGBM prediction model to predict default borrower in social lending platform

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ARTICLE INFO

Keywords:
Social lending
Credit risk prediction
Machine learning
Explainable AI
Recursive feature elimination

ABSTRACT

This paper proposes an explainable AI (XAI)-based prediction model utilizing the LightGBM algorithm to predict the likelihood of borrower default on a social lending platform. The dataset used in this study was obtained from Lending Club and consisted of various borrower characteristics and loan features. The proposed model not only provides high accuracy (0.87) in predicting defaulted borrowers, but also offers an explanation of the factors that contribute to the prediction. The model interpretability is facilitated through LIME and SHAP values, where SHAP values provide insights into the feature importance for the prediction. The outcome shows that the proposed model outperforms traditional approaches and delivers valuable insights for lending decision-making. The proposed model can be useful for lenders and regulators in the lending industry to improve decision-making processes and mitigating risk. Moreover, the XAI approach enables transparency and accountability in the decision-making process, making it more understandable and trustworthy for stakeholders.

1. Introduction

A social lending platform, also known as peer-to-peer lending or P2P lending, is an online marketplace that connects individual borrowers with individual lenders. These loans are made directly to borrowers and are often connected with non-profit organizations that provide microfinance (Bachmann et al., 2013). The process of borrowing and giving financial support through social lending can be faster than other forms of banking, particularly if the borrower lives in a remote location or if they need money quickly to turn their idea into a successful startup (Gao et al., 2021). Lenders can also get a better interest rate on the money they invest than they would get through a more traditional banking institution. Social lending works by combining technology, social networking, and microfinance to help borrowers in emerging countries gain access to loans on a local level (Wang et al., 2021; Wang & Ni, 2023). The process of how social lending operates is very straightforward: lenders and borrowers are matched together online through the microfinance company, with all lenders seeing the borrower's profile and using this information to decide who to lend money to Galema (2020). The lender gets together with the borrower and discusses an agreement that both parties are comfortable with, which is then recorded electronically by the social lending website such as Lending Club, Prosper, Zopa, and Kiva (Rao et al., 2020).

Unbalanced information among borrowers and lenders is a concern in social lending online. To put it another way, the lender does not have as complete an understanding of the borrower's reliability as the borrower does (Zwilling et al., 2020). This imbalance in knowledge might potentially lead to both adverse selection and moral hazard. In general, credit risk occurs when a lender or borrower in a debt arrangement fails to return the obligations entirely or in installments on the agreed-upon schedule. This might cause a delay in the full repayment of the obligation (Yeo & Jun, 2020). As a consequence, it is the effect of defaulting borrowers and lenders under derivatives arrangements. Today, academics from numerous fields recognize the significance of credit risk, and credit risk assessments have gotten a considerable amount of attention, especially in the financial services industry. To minimize lenders' financial risk, it is critical to forecast defaults and evaluate borrowers' creditworthiness. Bad loans threaten new investors, social lending companies, and borrowers. This highlights the utmost significance of the method for assessing borrowers and estimating the likelihood of default risk.

In modern days, technologies such as Machine Learning (ML),

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https://doi.org/10.1016/j.iswa.2025.200514
Received 8 July 2024; Received in revised form 25 November 2024; Accepted 26 March 2025
Available online 27 March 2025

Artificial Neural Network (ANN), and big data are being used extensively in risk management systems within the financial sector (Kim & Cho, 2019). Support Vector Machines (SVM), Random Forest (RF), Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), and Deep Neural Networks (DNNs), ANN, are some of the most popular methods that are used to forecast risk in social lending (Aleksandrova, 2021; Chen et al., 2019; Costello & Lee, 2019; Li et al., 2021b; Liu et al., 2023; Malekipirbazari & Aksakalli, 2015; Teply & Polena, 2020; Zanin, 2020). In the previous studies, Toply and Polena (2020) used six different ML models including ANN, K-Nearest Neighbors (KNN), SVM, LR, Linear Discriminant Analysis (LDA), and RF models to predict the possibility of default loan on a social lending platform using LendingClub data. The outcomes showed that the ANN, LDA, and LR models performed well in term of prediction accuracy. Similarly, Malekipirbazari and Aksakalli (2015) used RF, KNN, LR, and SVM to analyze the default borrowers on a social lending platform, where RF had high prediction accuracy.

Another study by Li et al. (2021b) used deep learning (DL) algorithms to predict loan defaults in social lending. The authors used a combination of convolutional neural networks (CNN) to pick the features from loan data and predict default probability. The experimental outcomes presented showed that the proposed model had higher accuracy and higher F1-score than traditional ML models. Furthermore, Aleksandrova (2021) applied numerous ML techniques such as DT, LR, gradient boosting machine, XGBoost, MLP, and RF to evaluate credit scores for lending platforms using the LendingClub dataset. This study's result shows that XGBoost performed well compared to other ML models. In another study, Chen et al. (2021) used resampling and cost-sensitive techniques to analyze unbalanced datasets and several ML approaches such as LR, neural networks (NNs) and RF to forecast social lending default risk. Zhu et al. (2019) applied the SMOTE method to solve the imbalance class problem for the dataset and build a RF based default prediction model. Zanin (2020) investigates the performance of the default prediction by class combination rebalancing methods with a variety of ML and regression approaches and analyzes the results. Moreover, in order to enhance the effectiveness of the prediction, they suggest combining numerous probability predictions.

Clustering techniques have been applied to group borrowers into different risk categories. For instance, Song et al. (2020) suggest using DM-ACME, or distance-to-model and adaptive clustering-based multi-view ensemble, to predict default risk on social lending platforms. In order to create an ensemble of varied ensembles made up of gradient boosting decision trees, this technique explores multi-view learning using an adaptive clustering method.

Risk assessment in online lending has been the subject of several researchers' studies. Guo (2020) built a BP neural network-based method for the evaluation of loan risk and compared it with LR. The experimental outcome favors the BP neural network-based approach over the conventional logistic regression technique. Li et al. (2021) and Pandey et al. (2018) used various classification techniques to analyze the credit risk for the lending platform. Ko et al. (2022) proposed a prediction model to evaluate the default risk on social lending platforms. This study used five AI models, such as LightGBM, DT, RF, CNN, and ANN, and three statistical models, such as LDA, LR, and the Bayesian classifier. In another study, Bussmann et al. (2020, 2021) also used Explainable Artificial Intelligence (XAI) for credit risk assessment. Bussmann et al. (2021) suggest an XAI model that is suitable for credit risk management, especially when assessing the risks involved in lending platforms. The proposed model uses networks of correlation in conjunction with Shapley values to group AI predictions based on the similarity of their underlying explanations. The author highlights the challenges of traditional risk management methods and argues that XAI can provide greater transparency and interpretability in the decision-making process. The authors applied logistic regression, XGBoost ML models and model-agnostic methods. Ariza-Garzón et al. (2020) applied logistic regression alongside the XAI-SHAP interpretation model to evaluate credit risk. By utilizing the LendingClub dataset, the study showed that ML credit scoring models can achieve both accuracy and transparency. Ohana et al. (2021) applied gradient boosting decision trees (GBDT) and DL model to predict S&P 500 price drops, with GBDT outperforming other models by using fewer key features. Shapley values identified important crisis variables, and the analysis of the March 2020 crash revealed the tech sector's predictive role. De Lange et al. (2022) introduced an XAI model designed to predict credit defaults using a distinct dataset of unsecured consumer loans from a Norwegian bank. They integrated a LightGBM model with SHAP, allowing for the interpretation of the explanatory variables influencing the model's predictions. Islam et al. (2023) employed 10 different ML and XAI methods to predict bank loan defaults. The study's outcome shows that RF outperformed other machine learning models. Shang et al. (2023) developed a cross-domain network for predicting credit defaults, termed Transfer Light Gradient Boosting Machine (TrLightGBM), which is based on an interpretable method of integration and transfer.

The absence of XAI in previous social lending research has hindered understanding of the ML models used, complicating informed decision-making for investors. As social lending grows in popularity, ensuring transparency and interpretability is essential. Incorporating XAI can enhance transparency and accountability, crucial for maintaining trust. This study employs XAI to analyze prediction results, helping investors build trust and make informed decisions by clarifying the factors influencing lending decisions. The purpose of this research is to construct prediction models for one of the biggest social lending platforms offered by Lending Club (https://www.lendingclub.com/) in the US so that investors may receive borrowers' credibility early, thereby increasing information transparency and decreasing default risk. In addition, investors may manage problematic loans in advance and minimize their losses before investing them. According to Ribeiro et al. (2016) lenders often find it difficult to trust new technologies and prediction models in social lending. To deal with this kind of problem, XAI is employed to analyze the model's prediction result. XAI results may help investors build their trust in the prediction model's outcomes.

The novelty of this study lies in its focus on enhancing transparency and interpretability in credit risk assessment for social lending platforms—an area where most previous research has concentrated on machine learning models without sufficient attention to explaining predictions. While studies like Guo (2020) and Ko et al. (2022) explored credit risk models, and Bussmann et al. (2020, 2021) and de Lange et al. (2022) applied interpretability methods in traditional banking, their use in social lending has been limited. To address this gap, we combine LIME and SHAP to interpret the LightGBM model and other machine learning models. This approach improves transparency and feature interpretation, helping investors understand key factors influencing credit decisions. Additionally, we applied Recursive Feature Elimination (RFE) to identify key features for social lending, further aiding investor decision-making. Using the LendingClub dataset from 2007 to 2020, this study delivers a comprehensive analysis of borrower credibility and default risk, contributing to the development of more accurate and interpretable credit scoring models.

The research question for this study is the following:

1. How does the explainable AI-based LightGBM prediction model compare to traditional ML models in predicting borrower default on social lending platforms?
2. What are the most important features used in the explainable AI-based LightGBM prediction model for predicting borrower default on social lending platforms?
3. How does the interpretability of the explainable AI-based LightGBM prediction model enhance the transparency and trustworthiness of lending decisions on social lending platforms?

2. Methodology

In this section, we will explain our methodology. In the methodology section, we have divided the process into seven steps, such as data preprocessing, sampling, data splitting, feature selection, building a ML model, evaluating the model based on evaluation metrics, and lastly applying XAI to explain the model results using SHAP and LIME methods. In Fig. 1, the research framework is shown.

The first step in our methodology is data preprocessing, where we clean and transform the raw data to remove any inconsistencies and errors. The second step is sampling, where we select a representative subset of the data to train and test the model. The third step is data splitting, where we split the data into training and testing sets to evaluate the model's performance. The fourth step is feature selection, where we identify the most relevant features that have a significant impact on the model's performance. The fifth step is building a ML model, where we train and test the model using the selected features. The sixth step is evaluating the model based on evaluation metrics, where we use metrics such as accuracy, precision, recall, and F1 score to assess the model's performance. Furthermore, we apply XAI techniques to interpret and explain the model's results using SHAP and LIME methods.

To illustrate our research framework, we have shown it in Fig. 1. Our methodology provides a systematic approach to solving complex problems, and it can be replicated by other researchers to build and evaluate ML models.

2.1. Data and preprocessing

The biggest social lending platform in the US, Lending Club, provided the data that we used in this paper. Lending Club dataset downloaded from Kaggle's website (https://www.kaggle.com/datasets/ethon0426/lending-club-20072020q1). The Lending Club has 2925,493 loan entries with 142 attributes. Based on the loan's duration, this data collection includes two types of loan records: 36 month and 60-month

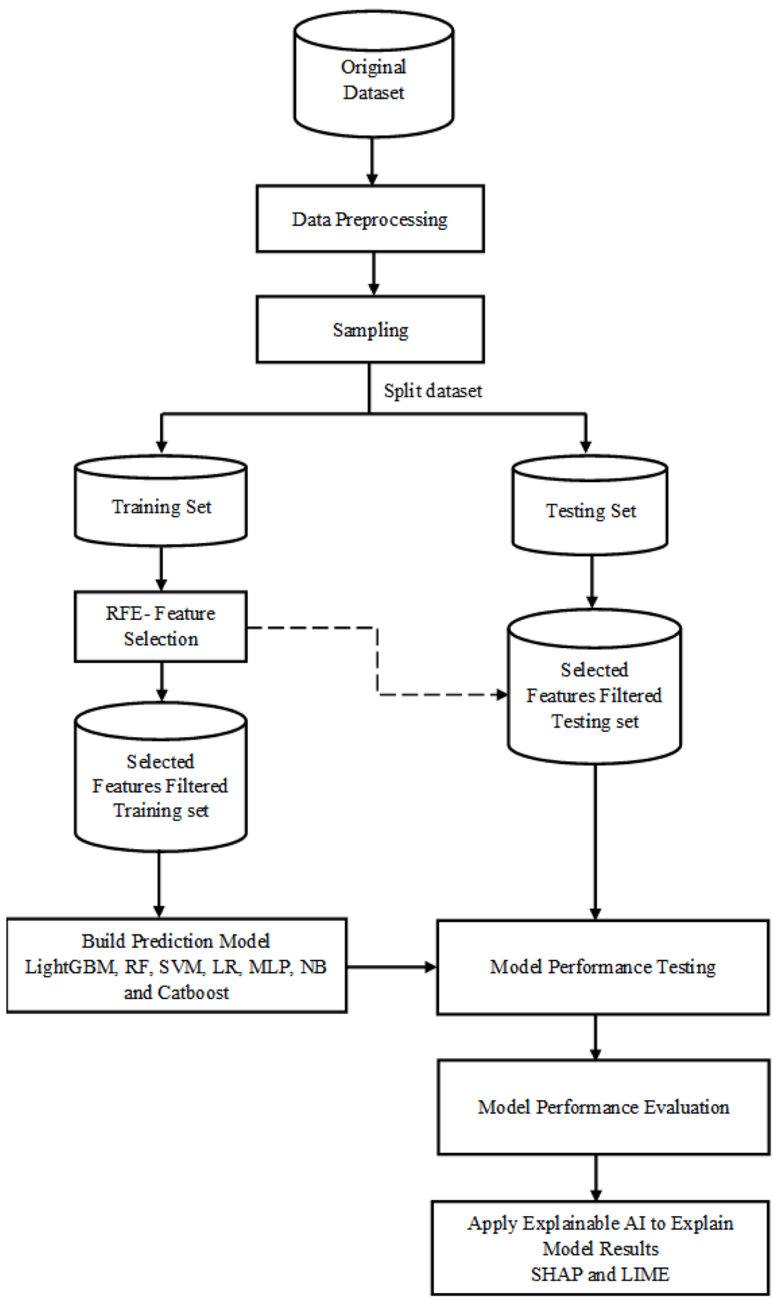


Fig. 1. Framework for the research.

loans. We have used a dataset from 2007 to 2020 to predict the defaulted borrowers. Before processing the model training data, we had to address the issue of missing values in the data. Firstly, we eliminated any features that had nothing to do with loans, like the borrower's loan ID, url, title, descriptive features, and empty and missing features, which have 80 % missing and empty values. After the deleted column, this study filtered 26 features based on previous studies (Emekter et al., 2015; Serrano-Cinca & Gutiérrez-Nieto, 2016; Yu & Zhang, 2021) Loan status, we consider to be target, has the values fully paid and charged off, respectively. In LendingClub, default occurs when an individual or organization fails to meet a financial obligation, typically by missing scheduled payments on a loan or bond. A loan is considered to be in default when payments are overdue for >120 days. Once a loan reaches default status, charge-offs are generally processed within 30 days.

In this study, fully paid is considered a non-defaulter, while charged off is considered a defaulter. The data required to be transformed into numerical forms because the majority of the data were presented as categories, which are unsuitable for model training. The target variable "loan status" is the first one. The "default category" was set to 1, and the "non-default" category was set to 0. The second one is "emp length," which denotes the number of years spent employed. We convert this value based on their years those with less than a year are listed as 0 and other numerical values range from 1 to 10 years. Other features which are values in categories, were also converted into numeric form.

In this study, we have applied max-min normalization to each column of the data. In an effort to improve the data's suitability for adoption by algorithms. Following are equation how normalization works:

$$X_{normalized} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

Using the normalization algorithm provided, all of the data for each lag are scaled to numbers between 0 and 1.

In Fig. 2, we analyze all the selected features based on loan status. Fig. 2(a) illustrates the loan status count, showing that non-default loans outnumber default loans. Fig. 2(b) presents the distribution of loan payments by loan status, indicating that most default and non-default loans have installment amounts between \$300 and \$600. In Fig. 2(c), the relationship between loan status and loan amount is depicted, highlighting that loans ranging from \$4000 to \$15,000 have the highest occurrence of both default and non-default cases. Fig. 2(d) shows loan status relative to loan grade, with grades B, C, A, and D being the most common, and grades C, D, and B having the highest number of default cases.

Fig. 2(e) illustrates the loan status in relation to the average current balance, with the majority of loans having an average balance of \$40,000. Fig. 2(f) shows loan status based on the debt-to-income (DTI) ratio, with most loan requests having a DTI of 40. Fig. 2(g) depicts the revolving utilization distribution by loan status, where the majority of loans have a revolving utilization between 40 and 160. Fig. 2(h) presents the distribution of interest rates based on loan status, showing that most loans have interest rates between 5 % and 25 %, and the default rate is notably higher when the interest rate falls between 10 % and 20 %. In Fig. 2(i), the loan status is shown alongside the last payment amount, revealing higher default rates when the last payment amount is 0. Finally, Fig. 2(j) illustrates the loan status in relation to revolving balance, with most loans having a revolving balance of \$100,000. Additionally, the distribution is concentrated near the lower end of the revolving balance spectrum, indicating that most loans, whether defaulted or not, have relatively low revolving balances. While the number of defaulted loans remains consistently smaller than non-defaulted loans, even at higher revolving balances, this suggests that revolving balance alone may not be a strong indicator of loan default. Overall, these graphs highlight the importance of key loan features in distinguishing default from non-default loans and offer valuable insights for refining predictive models.

Fig. 3 shows the distribution of loans from 2007 to 2020, categorized as Non-Default and Default. Loan issuance grew significantly from 2011, peaking in 2015 and 2016, where Non-Default loans were more frequent than Default loans. After 2016, both categories experienced a steady decline. Early years (2007–2010) saw minimal loans and defaults, but from 2013 to 2017, the proportion of Default loans increased, reflecting a higher default rate during the peak loan issuance years. The default rate, represented by the red line, shows a steady decline from 17.9 % in 2007 to a low of 12.6 % in 2010. It then increased sharply, peaking at 23.9 % in 2019, before plummeting to just 1.3 % in 2020.

2.2. Sampling

Data balancing is crucial for creating financial decision models since lending platforms often have vast that are unbalanced datasets. Class imbalance refers to datasets where the target classes' data distributions are uneven; for example, one class label could contain a lot of data while the other might have little data (Namvar et al., 2018). Fig. 2(a) highlights the significant class imbalance in the dataset, with 1497,783 Non-Default cases, representing 80.51 % of the total data after data preprocessing, and 362,548 Default cases, accounting for only 19.49 %. This imbalance could potentially impact the performance of ML models, as models may become biased toward predicting the majority class (Non-Default), making it more challenging to accurately predict the minority class (Default). Therefore, we employed the random-sampling strategy to choose balanced datasets so that we might use more balanced data to train our model.

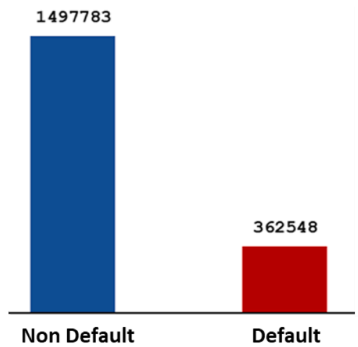
Random sampling is often chosen because it minimizes bias, ensuring every member of the population has an equal chance of selection. This approach enhances the generalizability of results, making findings more applicable to the broader population. Additionally, it's a straightforward and practical method to implement (Olken & Rotem, 1995). The most practical and straightforward way for selecting samples is the random sampling method. In this study for experimentation, we utilized a sample dataset of 20,000 records, of which 10,000 as "Fully Paid" and 10,000 as "Charged Off". The choice of 20,000 observations represents a balance between statistical rigor, computational efficiency, and the need to address class imbalance. This approach ensures that the model achieves robust performance on both Default and Non-Default classes while remaining practical and efficient. The selected dataset size is sufficiently large to capture the complexity of the problem while avoiding unnecessary computational overhead, ensuring reproducibility and scalability in real-world financial applications.

2.3. Training and test set

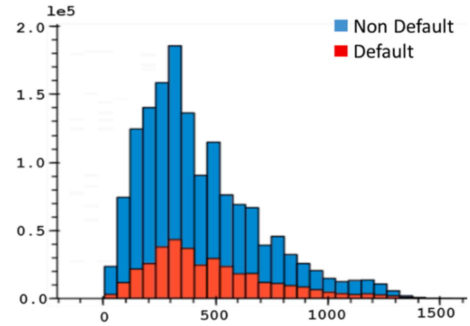
A model validation technique known as train-test-split enables you to simulate how a model might operate on new and untested data. In this study, the dataset was partitioned into two sets, namely the training and testing set. The training set was utilized for model training, while the testing set was reserved for evaluating the model's performance. We have used 80 % dataset for training set and 20 % dataset for testing dataset (displayed in Table 1).

2.4. Feature selection

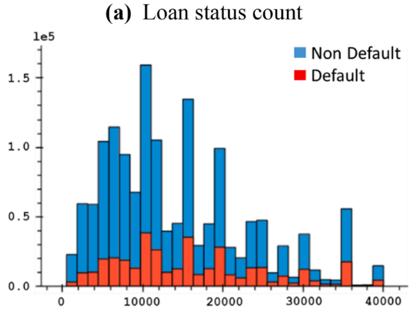
In this study, the RFE (Hou, 2020) method was used to identify significant features for social lending platforms. It can help the lender by identifying the default borrowers. RFE is a feature selection technique that fits an algorithm and eliminates the weak feature until the required number of features is attained. RFE aims to remove any potential collinearity and dependencies from the model by iteratively deleting a limited number of features every loop according to the model's coef_ or feature importances_ features. However, it is frequently unknown in advance how many features are legitimate, even as RFE stipulates a certain number of features to maintain. It automates the feature



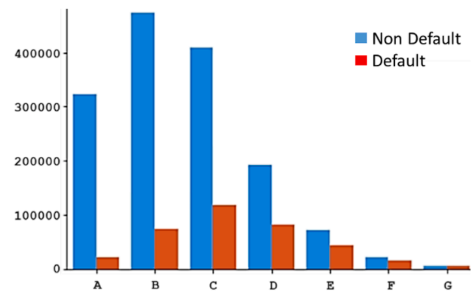
(a) Loan status count



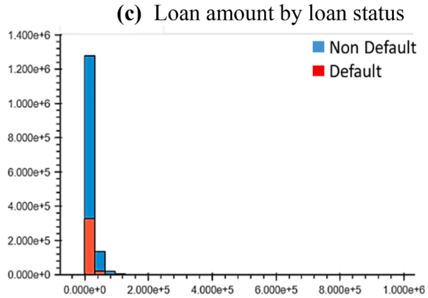
(b) Installment by loan status



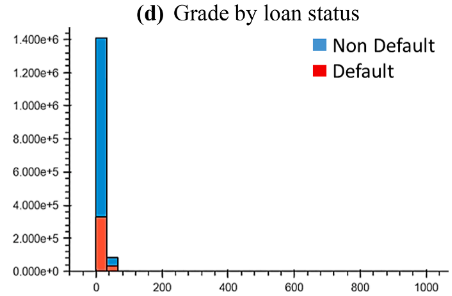
(c) Loan amount by loan status



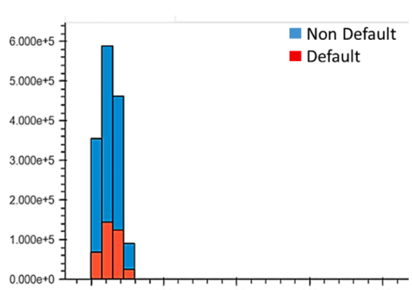
(d) Grade by loan status



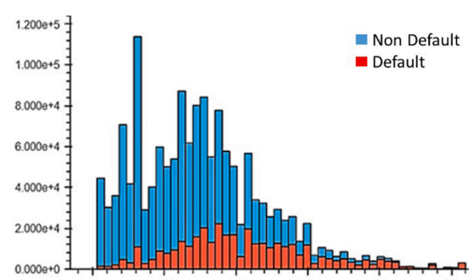
(e) Average current balance by loan status



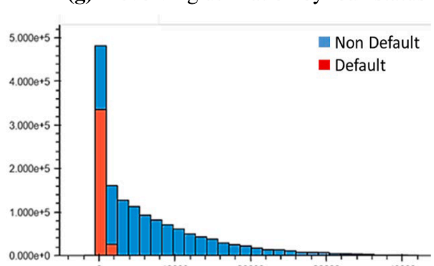
(f) DTI by loan status



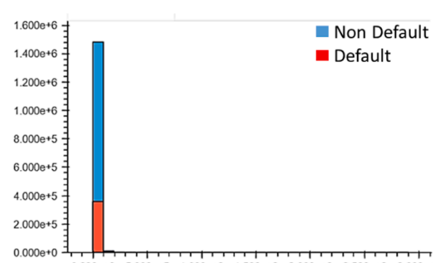
(g) Revolving utilization by loan status



(h) Interest rate by loan status



(i) Last payment amount by loan status



(j) Revolving balance by loan status

Fig. 2. Feature's data distribution based on loan status.

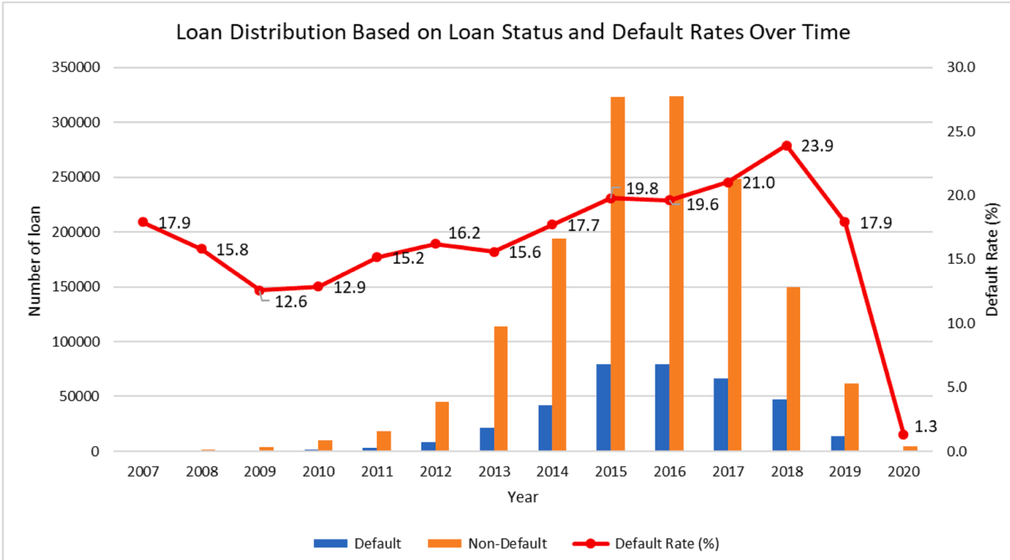


Fig. 3. Loan distribution based on loan status and default rate over time.

Table 1
Summary of data split.

Set	Numbers	Percentage (%)
Training	16,000	80 %
Testing	4000	20 %
Total	20,000	100 %

selection process, making it easier and more efficient, especially in models with a large number of features

To determine the ideal number of characteristics, RFE uses cross-validation to score many feature subsets and choose the finest scoring set of attributes. We have used RF-RFE. This study identified the top 10 features to build the ML models based on the RFE displayed in Table 2.

The algorithm for RF with RFE is displayed in the Algorithm. 1. The algorithm iteratively removes the features with the lowest importance scores until the desired number of features is reached. The model is then evaluated on the remaining features to determine whether its performance is fitting or not. If the performance is not fitting, the process is repeated with the next lowest importance feature until the desired performance is achieved or all features are eliminated.

2.5. Build machine learning models

In this study, we build the LightGBM model and compare it with MLP, RF, CatBoost, LR, and SVM. These models are selected for their broad application and diverse methodological approaches in ML. MLP provides a comparative perspective on neural networks, while RF and CatBoost, as ensemble methods like LightGBM, serve as suitable points of reference. LR, a commonly used statistical model, offers a simple and interpretable baseline for classification. SVM is included due to its effectiveness in handling high-dimensional feature spaces, particularly in scenarios with a high ratio of features to samples. NB is chosen for its computational efficiency and ability to handle high-dimensional data. We used Python 3.5.1 as a programing language and the Sklearn library to build the models and do the data analysis. This study used the Anaconda (https://www.anaconda.com/) platform and Jupyter notebook to write the Python code. All the experiments were executed on a Windows system with an Intel(R) Core (TM) i9 and 32 GB of RAM. We trained our proposed model after the data pre-processing, feature selection, and data partition.

Table 2
Selected features for models.

Features	Description
loan_amnt	The borrower's indicated loan request's total amount.
last_pymnt_amnt	Most recent total received for payments
avg_cur_bal	Current average balance throughout all accounts
int_rate	The loan's interest rate
installment	The amount that the borrower would need to pay per month if the loan were to start.
grade	Loan grade assigned by LC.
annual_inc	The borrower's self-reported yearly income given at registration
dti	Monthly debt payments in total, excluding split by the proposed loan amount and the mortgage payment
revol_util	Revolving line utilization rate
revol_bal	The whole revolving credit balance
loan_status	Status of Loan applications (Target)

2.4.1. Random forest

Random forest (RF) (Breiman, 2001) is an ensemble learning technique utilized for various tasks, including regression, classification, and more. During the training phase, it builds multiple decision trees (DTs), and the last prediction is made by taking the mean or mode of the predictions made by the individual trees. It is a type of supervised learning algorithm and a subset of DT algorithms. A DT is a structured tree that can be binary or non-binary. Each branch of the tree corresponds to the result of a feature attribute that spans across a range of values. Additionally, every leaf node of the tree stores a category. Each non-leaf node links to a test of a feature. The DT starts with a root node, tests the feature characteristics in the category to be categorized, and then selects branches based on their results until the leaf node is reached (Zhu et al., 2019). The category kept by the leaf node is considered the decision outcome. A RF is a group of DTs, where each tree is independent of the others. The Gini index is used as a selection measure to divide the attributes in the DT. In the equation provided, "*C*" represents the total number of classes, and the probability of selecting a data point with class "*i*" is represented by "*p(i)*".

$$G = 1 - \sum_{i=1}^{C} p(i)^2 \tag{2}$$

2.4.2. Logistic regression (LR)

Logistic regression (LR) (Chen et al., 2019a) is a method of statistical analysis that helps explain the connection among a number of different

Algorithm 1
Algorithm to construct RF-recursive feature elimination.

```
Input: training dataset, number of desired features(n)
Output: top-most important feature set (F`)
LC_FS ← LendingClub features after data pre-processing
td ← training dataset
function RFE (LC_FS, td)
    train a RF model on the training dataset(td)
    compute the feature importance scores for each feature
    remove the feature with the lowest importance score from LC_FS
    If count (LC_FS)! = 0
        RFE (LC_FS, td)
    end
    return F`
end
```

independent factors and a single dependent variable. When compared to the dichotomous nature of the dependent variable, the independent variables could be categorical or continuous (LaValley, 2008). It is employed to solve classification problems and determine the likelihood that a binary incident will happen. The logistic regression is represented by the following equation:

$$f(x) = \frac{1}{1 + e^{-x}} \tag{3}$$

In the above equation x represents dependent variable, $f(x)$ is representing output of prediction, e represents natural logarithm base.

2.4.3. Support vector machine (SVM)

A Support Vector Machine (SVM) (Cawley & Talbot, 2002) is a powerful and popular supervised ML algorithm that is used regression, classification, and outlier detection. In order to maximize the separation between the various classes in the dataset, the SVM algorithm seeks a hyperplane. The method seeks the best hyperplane to maximize the margin between classes, or the separation between the nearest data points for each class. The SVM approach is very beneficial when the data is not linearly separable because it may utilize a kernel function to shift the data into a higher-dimensional space where it becomes separable (Wang & Li, 2019). This makes SVMs more flexible and able to handle complex datasets. The SVM method is capable of working with high-dimensional data, is resistant to overfitting, and can work with non-linear decision boundaries.

2.4.4. Light gradient boosting machine (LightGBM)

The Light Gradient Boosting Machine (LightGBM) was created by Microsoft MSRA in 2017 as a rapid and effective gradient-boosted decision tree (GBDT) method with an open-source purpose (Niu et al., 2019). It enables effective parallel training and can be employed for a wide range of ML assignments, including regression, classification, and sorting (Ko et al., 2022). Unlike XGBoost, LightGBM uses a histogram to quicken the training process, decrease memory usage, and employ a well-planned growth scheme that includes depth limitations. The separation point is computed using Gradient-based One-Side Sampling (GOSS) by calculating the variance gain. Additionally, LightGBM grows its leaves vertically (leaf-wise), which reduces loss and increases the algorithm's efficiency and accuracy, as depicted in Fig. 4.

In the case of binary classification, LightGBM seeks to predict a target variable y that takes on one of two values (0 or 1). The algorithm uses a sigmoid function to transform the output of the model into a probability value range of 0 and 1, which can then be used to make a binary classification decision.

The equation for the LightGBM binary classification equation displayed is as follows:

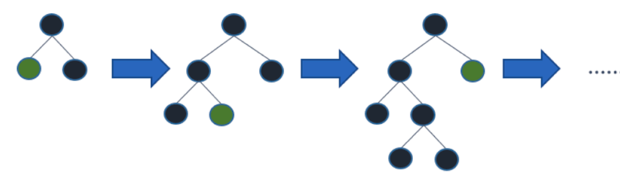


Fig. 4. LightGBM architecture.

$$F_m(x) = \sigma\left(\sum_{i=1}^{m} \gamma_i h_i(x)\right) \tag{4}$$

$F_m(x)$ is the prediction for the target variable y at the m-th iteration, given the input features x. σ is the sigmoid function, which transforms the output of the model into a probability value range of 0 and 1. γ_i is the learning rate or step size for the i th iteration. $h_i(x)$ is the weak learner or decision tree that is fit to the negative gradient of the loss function at the i th iteration. The overall goal of LightGBM is to find the optimal values of γ_i and $h_i(x)$ that minimize a specified loss function for the classification task. The algorithm uses gradient boosting to iteratively update the model parameters, by fitting the weaker learners to the negative gradient of the loss function at each iteration. Algorithm 2 is showing the pseudocode of LightGBM.

2.4.5. Multi-layer perceptron (MLP)

A multi-layer perceptron (MLP) is a type of feed-forward neural network that comprises three interconnected layers: input, hidden layers, and output. The input layer is where data input is defined and the signal to be processed is received (Chi et al., 2019). The output layer performs classification and prediction tasks. The MLP's computational engine contains an arbitrary number of hidden layers among the input and output layers, enabling it to estimate any continuous function and resolve nonlinear issues. MLPs are commonly used for recognition, prediction, and pattern classification (Gardner & Dorling, 1998). The equation for an MLP classifier can be expressed as follows:

$$z_j = \sum w_{ij} x_i + b_j \tag{5}$$

where z_j is the weighted sum of the inputs x_i from the previous layer, w_{ij} is the weight connecting the i th input to the j-th neuron, b_j is the bias term for the j-th neuron, and Σ is the sum over all the inputs.

2.4.6. Naïve Bayes (NB)

The Naïve Bayes (NB) classifier is an effective classification technique. It is easy to build and especially useful for extremely large data

Algorithm 2
Algorithm to build LightGBM.

```
Input: Training dataset Train_D = {(x^1, y^1), (x^2, y^2), ... (x^n, y^n)}, Testing_X dataset Test_X= {(x_t^1),
(x_t^2), ... (x_t^n)}, Testing_y dataset Test_Y = {(y_t^1), (y_t^2), ... (y_t^n)},
Output: Predict default borrower D
lgbm ← LightGBM
p← parameters {n_estimators=250, learning_rate=0.3, max_depth=2, reg_alpha=4,
reg_lambda=5, min_data_in_leaf=20}
Training
    | Train a lgbm model using the Train_D and p
end
Predict
    | Test the lgbm model using the Test_X dataset
    | Return D
end
Evaluate Model Performance
    | Calculate the accuracy, precision, recall and F1-Score of the model using Test_Y dataset and
    | D
End
```

sets. In addition, NB has been shown to surpass even the most advanced classification systems. The NB classifier relies on the Bayes theorem and makes independent predictions for each variable (Qasem & Nemer, 2020). A NB classifier, to put it simply, believes that the existence of one attribute in a class has nothing to do with the existence of any other attributes. Conditional probability is used in the Bayes theorem. (The possibility of an event happening if another event occurs is known as conditional probability. Eq. (5) represents NB utilizing the conditional probability.

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \tag{6}$$

Above

- *P(c|x)*: The posterior probability of a class (target c) given a predictor (attributes x).
- *P(c)*: Previous class probability.
- *P(x|c)*: The probability of a predictor in a particular class, or likelihood.
- *P(x)*: The probability of the predictor prior to observation.

2.4.7. Catboost

Catboost (Bentéjac et al., 2021) is a ML algorithm developed by Yandex that was released into the open-source community not too long ago. “Catboost” is an acronym that stands for “Category” and “Boosting.” It’s simple to combine with popular deep learning packages like TensorFlow from Google and Core ML from Apple. Its adaptability to a broad variety of data formats makes it useful for addressing several issues that modern enterprises must deal with. When compared to other top ML algorithms, Catboost consistently produces results at or near the state of the art. To quantify categories, we may utilize Catboost without any prior explicit processing (Zhou et al., 2021). Catboost uses a variety of statistical measures on combinations of categorical characteristics and combinations of categorical and numerical information to produce numerical representations of categorical values.

2.5. Explainable AI (XAI)

XAI is a collection of techniques for making ML’s findings and output more accessible and reliable to human users. XAI is employed to explain AI and ML models’ effects and possible biases (Dikmen & Burns, 2022; Miller, 2019). XAI’s primary objective here is to provide the model with as much interpretability as possible, which will, in turn, make it easier to analyze the trustworthiness of the ML model as well as the causality of its components. Generally, XAI systems evaluate the elements that went into the models and isolate the ones that were most influential. Here, we detail the most advanced post-hoc XAI methods currently available (Miller, 2019). In our opinion, it is critical to examine various post-hoc explainability methods to check whether the outcomes provide complete transparency of choices for the default risk prediction issue. In this paper, we employ SHAP for global explanations and LIME for instance-level explanations of model predictions. LIME and SHAP were selected as the explainability methods in this study due to their complementary strengths in providing both local and global interpretability, addressing the need for transparent and understandable ML models. XAI techniques, such as LIME and SHAP, aim to enhance the accessibility and reliability of ML model outputs for human users, making it easier to assess the model’s trustworthiness and identify any biases (Sahay et al., 2021; Jiang et al., 2023).

2.5.1. LIME

LIME (Locally Interpretable Model-Agnostic Explanations) is a ML interpretability technique used to explain the predictions made by a black-box model. It is a model-agnostic method, meaning that it may be applied to any ML algorithm regardless of its architecture or complexity. Ribeiro et al. (2016) claim that model-heretic explanations may be applied to explain any classification model, regardless of the method being used to make predictions, since LIME is unrelated to the prior classifier. Last but not least, LIME operates on a local level, which, in essence, implies that it is an observation-particular system. Moreover, much like SHAP, it will provide explanations for each and every individual observation that it collects. When it comes to the approach, the way that LIME acts are that it makes an attempt to build a local model by using sample data points that are comparable to the instance that is being discussed. The local model may belong to the category of possibly interpretable models, which includes decision trees, linear models, etc. These are the LIME-derived explanations for each observation x.

$$interpretation(x) = arg\ min_{v \in V}\ L(u,\ v,\ \pi_x) + \omega(v) \tag{7}$$

where v is the category of possibly interpretable models, which includes decision trees and linear models.

$v \in V$: A description that serves as a model.
u: being described is the primary classifier.
$\omega(v)$: Evaluation of the degree of explanatory complexity $v \in V$.

The proximity measure π_x determines the size of the region around the sample x, and this is what we take into account while providing an explanation. The important feature of LIME is that it is model agnostic; hence the purpose is to decrease the locality aware L loss without assuming anything about u. L is a metric for how inaccurately v approximates u in the region denoted by π_x.

2.5.2. SHAP

SHAP (SHapley Additive exPlanations) values, rooted in cooperative game theory, offer a method to explain the contribution of each feature in a ML model. The calculation of SHAP values is based on the Shapley value formula, which fairly distributes the total gains among players according to their contributions (Lundberg & Lee, 2017). The Shapley value for a feature i is given by the Eq. (7):

$$\varnothing_i = \sum_{S \subseteq N} \frac{|S|!(|N| - |S| - 1)!}{|N|} (f(s \cup \{i\}) - f(s)) \tag{8}$$

Here, N represents the set of all features, S is any subset of N not containing i, $f(S)$ is the model's prediction based on subset S, $|S|$ denotes the number of features in S, and $|N|$ is the total number of features. This formula sums the marginal contributions of feature i across all possible subsets S, weighted by a factor that ensures a fair distribution. The computational intensity of this process, due to the need to evaluate the model on all feature subsets, often leads to the use of approximation methods like Kernel SHAP or Tree SHAP, especially for tree-based models. In practice, a positive SHAP value indicates that a feature increases the model's prediction, while a negative SHAP value indicates a decrease, with the magnitude reflecting the strength of this contribution. Thus, SHAP values provide a comprehensive understanding of each feature's impact on the model's decisions, facilitating interpretability and transparency (Le et al., 2022).

2.6. Evaluation metrics

The five components of the confusion matrix are used to derive various metrics for assessing model performance. In this study, AUC, precision, recall, F1-score, G-mean and accuracy were employed to analyze the model's effectiveness. The confusion matrix elements are as follows: TP (True Positive) represents the accurately predicted quantity of "Default" loans; FP (False Positive) indicates the inaccurately predicted quantity of "Default" loans; TN (True Negative) denotes the accurately predicted quantity of "Fully Paid" loans; and FN (False Negative) corresponds to the inaccurately predicted quantity of "Fully Paid" loans. (see Table 3).

The accuracy of a binary classification issue is the performance measure that is used the vast majority of the time, and it may be computed using the equation.

$$accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{9}$$

Precision measures the quality of correct predictions, calculated as true positives over total positive predictions. Recall is the proportion of actual positives correctly identified, calculated as true positives over true positives plus false negatives. The F1-score combines recall and precision using their harmonic mean. The G-mean evaluates models with imbalanced data. Precision, Recall, F1-score, and G-mean are calculated as follows:

$$precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

Table 3
Confusion matrix.

	Actual "Fully paid"	Actual "Default"
"Fully Paid" Predicted	True Positive (TP)	False Positive (FP)
"Default" Predicted	False Negative (FN)	True Negative (TN)

$$F1\ score = \frac{2 * recall * precision}{recall + precision} \tag{12}$$

$$G - mean = \sqrt{precision * recall} \tag{13}$$

3. Experimental result

3.1. Model parameter

The hyperparameter tuning process for each model was conducted using a grid search, as detailed in Table 4. RF, LR, SVM, LightGBM, MLP, NB, and CatBoost were systematically optimized to identify the best configurations for each model's performance. Table 5 presents the parameter configurations for the models used in this study. Each row corresponds to a distinct model, with columns detailing the specific parameters and their respective values. The GridSearchCV method from Scikit-learn was employed to systematically identify the optimal parameter values from a predefined search space. These parameter settings were selected to maximize the predictive performance of each model, with the final configurations derived through an iterative process of experimental evaluation and refinement.

3.2. Classification results

Fig. 5 shows the ROC curve results for all the models, where LightGBM and Catboost both have a similar and highest AUC value: 0.94, RF's AUC value: 0.93, LR and MLP have similar AUC values of 0.91, and NB and SVM have AUC values of 0.89 and 0.88, respectively.

The classification result is shown in Table 6, where LightGBM performed better than other ML models based on performance metrics. In Table 6, the outcomes show that LightGBM has the highest accuracy (0.87), RF and CatBoost have similar accuracy (0.86), MLP has 0.84 accuracy, LR and SVM have similar accuracy (0.83), and NB has the lowest accuracy (0.81). When we examine the models' precision values, LightGBM and Catboost are comparable (0.82), whereas NB has the lowest (0.73). The maximum recall value is achieved by NB (0.96), followed by LightGBM, Catboost, SVM have 0.94, and MLP (0.93). Additionally, when comparing F1-scores, LightGBM and Catboost have the greatest value of 0.88, while NB has the lowest Recall value of 0.83. This study also calculates the G-mean score, where LightGBM and Catboost have the highest G-mean scores and NB has the lowest score.

According to the classification findings, LightGBM and Catboost performed well, however, when comparing accuracy and model runtime, LightGBM is superior for identifying defaulted borrowers.

3.3. Explainable AI results

3.3.1. SHAP

The features are presented in Fig. 6(a) in an order that goes from

Table 4
Hyperparameter tuning grid for model optimization.

Model	Parameter
RF	n_estimators= [50, 100, 200], max_depth= [10, 20, None], min_samples_leaf= [1, 5, 10], max_features= [1, 2, 3, …, 10], random_state=10
LR	C= [0.1, 1, 10, 100], solver= ['lbfgs', 'liblinear']
SVM	C= [0.1, 1, 10, 100], kernel= ['linear', 'rbf', 'poly'], random_state=10
LightGBM	n_estimators= [100, 250, 500], learning_rate= [0.01, 0.1, 0.3], max_depth= [3, 5, 7], reg_alpha= [1, 4, 10], reg_lambda= [1, 5, 10], random_state=10
MLP	hidden_layer_sizes = [(5,2), (10,5), (50,25)], alpha= [1e-5, 1e-4, 1e-3], solver= ['lbfgs', 'adam']
NB	verbose=1, cv=10, n_jobs=−1
Catboost	iterations= [500, 1000], learning_rate= [0.1, 0.3, 0.5], depth= [6, 8, 10], l2_leaf_reg= [3, 5, 10], random_state=10

Table 5
Model's parameter.

Model	Parameter
RF	bootstrap=True, n_estimators=100, min_samples_leaf=10, max_depth=20, criterion="gini", max_features=8, random_state=0
LR	penalty = 'l2', solver = 'lbfgs', random_state = 1, $C = 10$
SVM	$C = 10$, kernel = 'rbf', random_state = 0
LightGBM	n_estimators=250, learning_rate=0.3, max_depth=3, reg_alpha=4, reg_lambda=5, min_data_in_leaf=20, num_iterations = 500, random_state=0
MLP	solver= 'lbfgs', alpha=1e-5, hidden_layer_sizes= (5,2), max_iter=300, random_state=1
NB	verbose=1, cv=10, n_jobs=−1
Catboost	iterations = 1000, depth = 10, learning _rate=0.5, l2_leaf_reg=5, random_state=0

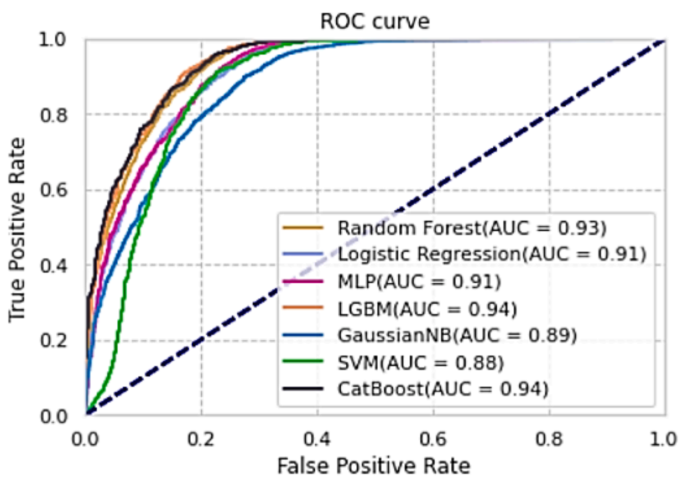


Fig. 5. ROC curve.

those having the greatest impact to those having the least impact on the prediction. Since it takes into consideration the absolute SHAP value, it is unrelated whether the feature has a negative or positive impact on the prediction. Based on the model, the last payment amount has the strongest predictive potential. The loan amount has the second-highest predictive influence among the features. The interest rate is the third feature that has the highest potential for prediction, and grade has lowest predictive influence among the features.

The list of significant features is shown in Fig. 6(b), with the most relevant feature listed first and the least relevant feature listed last (from top to bottom). Since each of the colors occupies half of the rectangles, it would seem that all of the features contribute the same amount to either a class being predicted by default (class = 1) or not (class = 0). Based on the model, the last payment amount has the strongest predictive potential. The loan amount has the second-highest predictive influence among the features. The interest rate is the third feature that has the highest potential for prediction.

The Fig. 7 illustrate the impact of various features on a ML model's predictions, likely for financial outcomes such as loan approvals. Fig. 7 (a) is a SHAP waterfall Plot, showing the influence of each feature on the model's output. Positive SHAP values (red) increase the prediction, while negative values (blue) decrease it. The most significant features include loan_amnt and int_rate, both negatively impacting the prediction, and last_pymnt_amnt, which has a substantial positive influence. Other features like avg_cur_bal and installment have moderate effects, while grade and dti show minimal impact.

Fig. 7(b) is a decision plot that shows how the model's prediction is constructed incrementally by considering each feature. The line plot traces the cumulative effect of each feature on the final prediction, starting with significant negative contributions from loan_amnt and int_rate. The last_pymnt_amnt notably boosts the prediction positively. Subsequent features add smaller refinements to the final output. Together, these visualizations highlight how individual features drive the model's predictions and their relative importance in determining the outcome.

Table 6
Model classification results.

Models	AUC	Accuracy	precision	recall	F1-score	G-mean	Runtime (Sec.)
RF	0.93	0.86	0.81	0.94	0.87	0.87	4.436
LR	0.91	0.83	0.79	0.93	0.85	0.86	0.328
SVM	0.88	0.83	0.78	0.94	0.85	0.86	9.714
LightGBM	0.94	0.87	0.82	0.94	0.88	0.88	0.328
MLP	0.91	0.84	0.79	0.93	0.86	0.86	1.150
NB	0.89	0.81	0.73	0.96	0.83	0.84	0.250
CatBoost	0.94	0.86	0.82	0.94	0.88	0.88	4.069

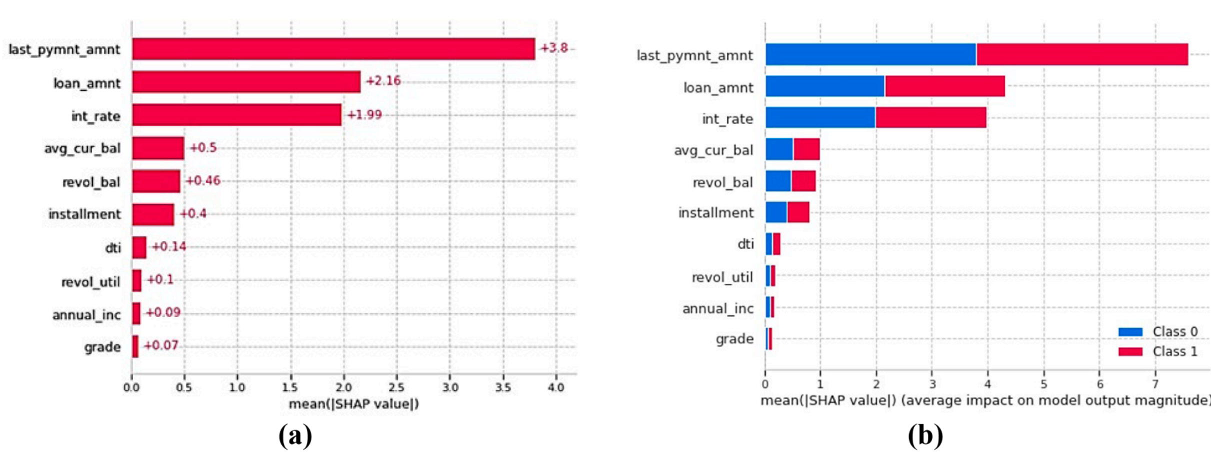
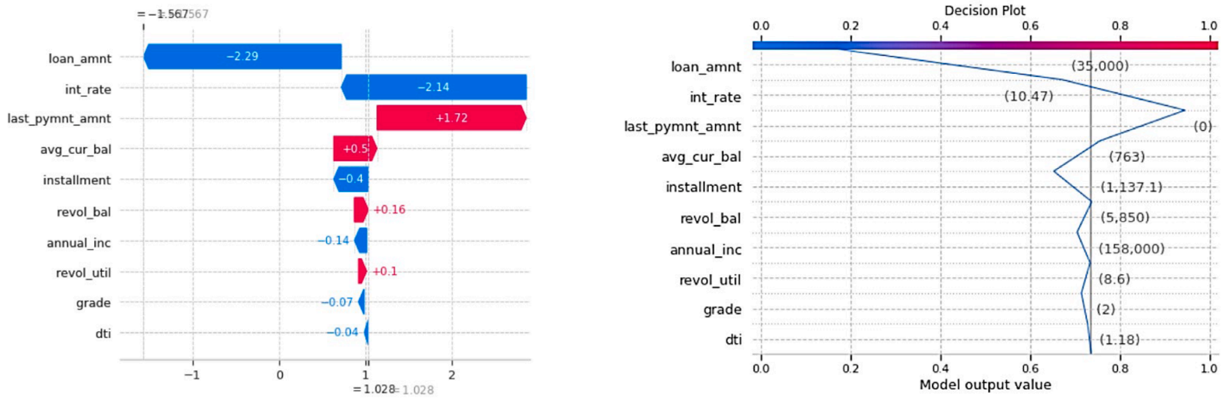


Fig. 6. **(a)** The light-GBM model's SHAP variable significance plot **(b)** Important features Plot.



(a) **(b)**

Fig. 7. (a) Waterfall plot **(b)** Decision plot.

Fig. 8 provides us with information on the explainability of a single prediction of a model. In the context of error analysis, the force plot may be used with the intention of elucidating the thought process that went into developing a specific case prediction. In this figure, the negative SHAP values are shown on the left and the positive values are shown on the right, as if they were in competition with one another. In force plot, −3.62 value is model output. The number that is referred to as the base value is what would be expected if we were unaware of any feature specific to the present instance. The base value is determined by taking the model's output throughout the whole training dataset and averaging it. The values on the figure arrows correspond to the feature's value in this particular case. In Fig. 8, the avarage current balance is $763.0 and the loan amount is $35,000.0 with a 10.47 % interest rate. Red indicates features that increased the model's score, whereas blue indicates traits that decreased the score. The effect of the feature on the result increases with arrow size. The x-axis allows us to see how much the impact has increased or decreased.

3.3.2. LIME

In the Fig. 9, the confidence level of the model's prediction is shown on the left side of the figure. The top 10 features that the model considered are shown on the right side. Finally, these identical features are displayed in the center, along with their contributions to directing the model's predictions toward the class. We provide a sample of a loan contract that relates to Borrower A and that the LightGBM model properly anticipated. This explanation may be interpreted as follows:

Borrower A is categorized as "default," indicating an inability to repay the full loan. The model shows a 92 % confidence level in this prediction. As illustrated in Fig. 9, key factors contributing to Borrower A's default include loan amount, interest rate, DTI, average current balance, grade, and annual income. XAI analysis reveals that loan amount, interest rate, and last payment amount have the highest impact on loan prediction due to their direct relation to financial risk. A larger loan amount increases the lender's risk and the likelihood of default. The interest rate affects the loan's profitability; a higher rate increases lender earnings but also raises borrower costs. Lastly, the last payment amount provides insights into the borrower's financial stability, where consistent large payments indicate financial health, while late or small payments suggest potential financial struggles.

4. Discussion

In recent years, individuals have increasingly turned to social lending platforms as alternatives to traditional banks and credit unions. The success of these platforms depends on the accuracy and reliability of ML models used to assess borrowers' creditworthiness. Transparency is crucial, as a lack of it can lead to concerns about trust and fairness. This study proposed an XAI-based LightGBM prediction model for predicting default borrowers on social lending platforms in comparison with other ML techniques such as LR, RF, MLP, SVM, NB, and CatBoost. The LightGBM model was more accurate and interpretable, thus being a valuable tool for lenders to use in making informative decisions and effectively taking precautions.

The model achieved the highest AUC (0.94), accuracy (0.87), precision (0.82), F1-score (0.88), and G-mean (0.88) of all models, with a runtime of 0.328 s. Furthermore, the performance of RF and CatBoost was close to LightGBM, which can be attributed to the shared ensemble learning structure. Ensemble algorithms, such as RF, LightGBM, and CatBoost, aggregate predictions from multiple base models, thereby enhancing robustness and reducing variance (Omer & Shareef, 2022). Additionally, these models efficiently handle categorical variables and large datasets, with comparable strategies for managing model complexity, mitigating overfitting, and identifying patterns in hierarchical data, contributing to their similar performance across diverse applications.

The ten most important features influencing default prediction were identified as: last_pymnt_amnt, avg_cur_bal, loan_amnt, int_rate, grade, installment, annual_inc, dti, revol_util, and revol_bal. These features

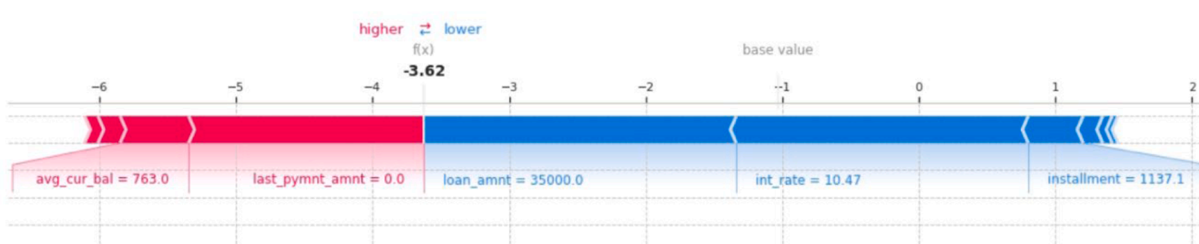


Fig. 8. Force plot.

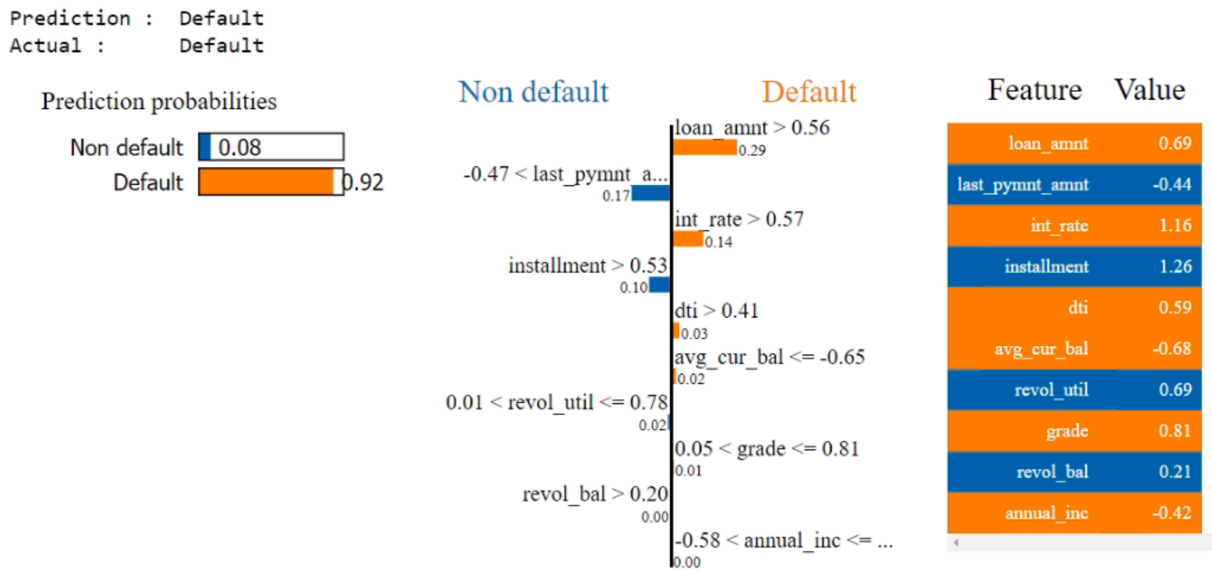


Fig. 9. LIME explanation for a borrower categorized as a "Fully paid" loan type by LightGBM model.

help set a focused framework for assessing borrower risk. To enhance the transparency of the LightGBM model, SHAP and LIME were employed, providing greater clarity and trust by explaining how decisions were made, detecting bias, and holding the model accountable. The XAI-based LightGBM model developed in this research offers not only accurate predictions but also interpretable explanations, ensuring transparency in decision-making on social lending platforms.

In comparison to the study by de Lange et al. (2022), which introduced an explainable artificial intelligence (XAI) model using LightGBM with SHAP for predicting credit default on a dataset from a Norwegian bank, our research presents several notable advancements. While their model exclusively employed SHAP for interpretability, we integrated both SHAP and LIME to enhance the explanatory power of the predictions. Additionally, while their analysis focused on factors like credit balance volatility and customer relationship duration, our study utilized the LendingClub dataset, allowing for a broader application of XAI methodologies across various financial sectors. Notably, their LightGBM model achieved an accuracy of 0.82, whereas our proposed LightGBM model demonstrated a superior accuracy of 0.87. The incorporation of both SHAP and LIME in our research provided deeper insights into the factors influencing defaults, offering a more comprehensive approach to improving model transparency and interpretability in financial risk assessment.

In the future, our goal is to enhance the model by incorporating additional elements like credit histories and customer consumption patterns, which will boost its accuracy and flexibility. We're also looking at adopting more sophisticated models, including various deep learning and ensemble learning approaches, to enhance the model's predictive capabilities. Furthermore, we aim to broaden the application scope of our model. While the primary focus of this study has been on predicting default risk in social lending, this approach shows potential for application in other domains, such as intrusion detection and risk prediction in the healthcare sector.

5. Conclusion

This study underscores the importance of integrating explainable AI (XAI) with classification methods, highlighting the introduction of the LightGBM model. This model effectively predicts default borrowers on social lending platforms, providing both accuracy and interpretability. By employing XAI, the study ensures that the decision-making process is transparent, addressing potential concerns about trust and fairness in automated credit assessments. Furthermore, the study identifies critical features that significantly impact the prediction of default borrowers. These features, derived using the RFE method. This detailed feature analysis equips investors with valuable insights, enabling them to make informed lending decisions.

The inclusion of XAI not only enhances the model's transparency but also provides actionable insights into the factors influencing borrower defaults. This dual benefit of accuracy and interpretability makes the LightGBM model a powerful tool for investors on social lending platforms. The study's findings pave the way for future enhancements, such as incorporating additional data elements like credit histories and customer consumption patterns, and exploring advanced ML techniques. This approach ensures that the model remains robust, flexible, and applicable to broader risk prediction domains, such as intrusion detection and healthcare risk assessment.

This study has several limitations that may affect its effectiveness and generalizability. Despite efforts to balance the dataset, it remains imbalanced, which could impact the model's performance in real-world scenarios. Additionally, the model's focus on Lending Club data limits its applicability to other platforms, and the absence of external validation reduces its robustness. These limitations highlight areas for future research to improve the model's accuracy and applicability. Future studies will explore advanced neural network techniques, such as deep learning, GANs, and NLP applied to platform conversations, to identify default borrowers and uncover new patterns in the data. Moreover, we will apply other interpretable machine learning (IML) methods to improve the transparency of machine learning models. The incorporation of high-performing deep learning models, in combination with IML approaches, will further enhance predictive accuracy and model interpretability in credit risk prediction.

CRediT authorship contribution statement

Li-Hua Li: Writing – review & editing, Conceptualization, Supervision, Validation, Resources. **Alok Kumar Sharma:** Conceptualization, Methodology, Formal analysis, Software, Writing – original draft, Visualization, Investigation. **Sheng-Tzong Cheng:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study is publicly available on Kaggle and can be accessed at https://www.kaggle.com/datasets/ethon0426/lending-club-20072020q1.

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